

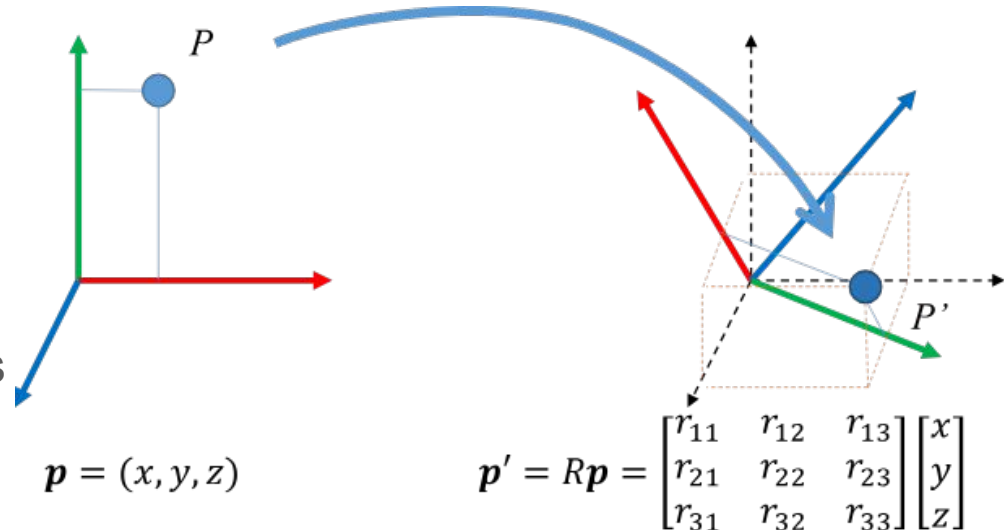
Previously on IA368 II



ROS2

TF - what is a transformation ?

- A mathematical operation that changes a point's coordinates from one coordinate system to another
- In robotics, the positions and orientations of objects (e.g., robots, sensors, and obstacles) are often expressed as transformations of one coordinate frame into another
- As time advances, these transformations change (e.g., if the robot moves)
- Although the choice of frames is arbitrary, the right choice of frames can greatly simplify the math required



TF - what is a transformation ?

- ROS provides a library called **tf2** for tracking these frames as they evolve in time
- **tf2 maintains the relationship between coordinate frames in a tree structure buffered in time** and lets the user transform points, vectors, etc. between any two coordinate frames at any desired point in time
- Implemented as publisher/subscriber model on the topics **/tf** and **/tf_static**
- For transformations, we call **broadcaster/listener** model
- There is a [message](#) for transformations:

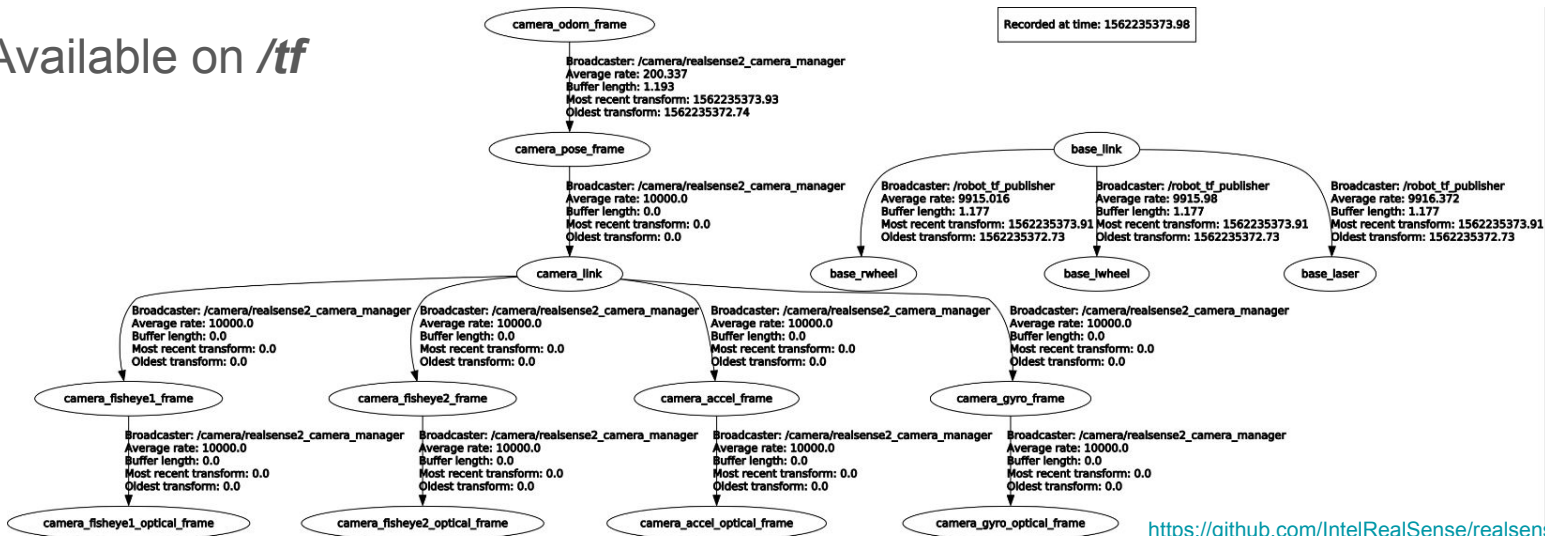
```
geometry_msgs/TransformStamped[] transforms
std_msgs/Header header
  uint32 seqtime stamp
  string frame_id
  string child_frame_id
geometry_msgs/Transform transform
  geometry_msgs/Vector3 translation
  geometry_msgs/Quaternion rotation
```

TF - what is a transformation ?

- A [broadcaster](#) publishes frames to */tf*
- The transform converts **frame_id** into **child_frame_id**
- A [listener](#) subscribes to */tf* and maintains a time-history of the received frames
- The buffer can then be queried for information about the transforms
- Some frames do not change: these are known as **static frames**
- [Static frames](#) can be broadcast to the */tf_static* to consume less bandwidth
- tf2 stores a time snapshot for every transform (for up to 10 seconds by default)
- When working with transformations, you can use the **last available** transform, or use a transform at a **specific [time](#)**

TF Tree - what is a transform tree ?

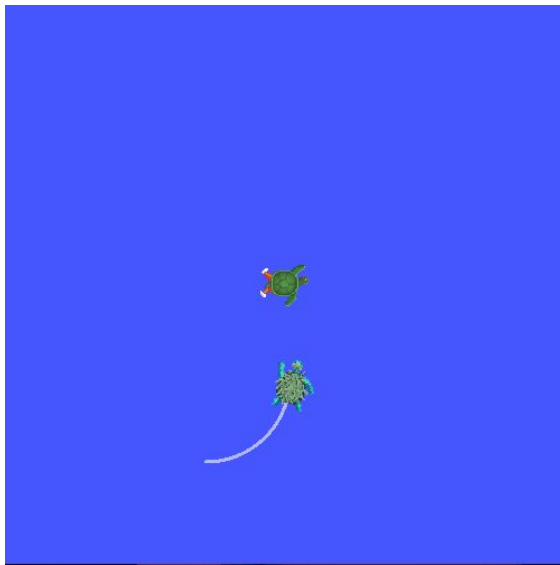
- In ROS, the transforms form a tree, with every node corresponding to a frame
- Every frame has one parent and an unlimited number of children
- Every frame has a unique name, called a **frame_id**
- Every transform on the tree is **timestamped**
- Available on `/tf`



Turtle time !

Let's practice with turtlesim:

<https://docs.ros.org/en/humble/Tutorials/Intermediate/Tf2/Introduction-To-Tf2.html>



TF - tools

- From the example, `ros2 run tf2_tools view_frames` to generate the TF tree graph in pdf file and `ros2 run tf2_ros tf2_echo [source_frame] [target_frame]` to report the transform between two frames
- Since `/tf` is just a topic, the TF tree is available using `ros2 topic echo /tf`
- It's possible to see the TF tree graph in real time using `ros2 run rqt_tf_tree rqt_tf_tree` or just `rqt`
- Instead of a pretty big command to use Rviz, just type `rviz2` and configure it manually
- You can create a static tf2 frame using `ros2 run tf2_ros static_transform_publisher`

TF - words of wisdom

- You can try to scan for failures in the tf with `ros2 doctor`
- Timing is crucial: if two frames came in too far apart, tf will throw an exception rather than interpolating
- Make sure your transforms form a connected tree, i.e., no disconnected components
- There are no default frames, only frames that you add
- Probably, you're not the first one facing problems with tf, so check for common errors from the community, like: <https://wiki.ros.org/tf/Errors%20explained>
 - TF_OLD_DATA is pretty annoying, this can be caused by restarting a bag playback (a bag in loop) or restarting a simulator

Thank You...

...for your attention

